Negative Group Delay Circuits: A Trade-Off Driven Review of Modern Design Strategies

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***Abstract*—Negative Group Delay (NGD) circuits are widely used in current high-frequency communication systems for delay equalization, dispersion correction, and timing alignment. The literature describes a range of NGD designs, including passive, active, reconfigurable, compact, and high-frequency implementations. Nevertheless, the performance of these methods is typically restricted by the presence of conflicting design requirements, which impairs the direct comparison of various approaches and the identification of common design trends.**

This study provides a trade-off analysis of selected state-of-the-art NGD circuits. Instead of classifying NGD circuits according to their implementation technology, the surveyed studies are comparatively examined using five recurring design trade-offs: bandwidth versus group-delay flatness, NGD magnitude versus insertion loss, passive stability versus active tunability, compactness versus performance, and high-frequency integration challenges.

The review shows that significant progress has been achieved using equal-ripple synthesis, reconfigurable tuning, active compensation, compact transmission-line structures, and integrated high-frequency realizations. However, no architecture currently achieves an optimal trade-off among all performance objectives. The analysis highlights the remaining challenges and provides an organized overview of current NGD design trends for future research and development.

***Index Terms*—Negative Group Delay (NGD), Non-Foster Circuits, Group Delay Flatness, Wideband NGD, Group Delay Dispersion.**

I. Introduction

The rapid development of 5G, 6G, and other high-frequency communication systems has increased the demand for high-speed signal processing with minimal distortion. Group-delay dispersion in these systems can degrade signal integrity and increase the bit-error rate. Therefore, NGD circuits have attracted considerable attention for delay equalization, dispersion compensation, and timing alignment.

NGD is often mistaken for a non-causal phenomenon. In practice, it can be explained through wave-interference mechanisms in resonant structures and active networks [6]. Early NGD implementations were closely related to Non-Foster circuits employing negative capacitors and inductors, although such approaches often suffer from stability and implementation challenges [4].

Current research on NGD circuits is typically categorized into two topologies: passive and active. However, considerably less attention has been given to examining how fundamental design limitations influence practical implementation trade-offs. To address this issue, this review compares representative NGD studies covering analytical synthesis methods, hardware designs, reconfigurable architectures, compact implementations, and high-frequency systems from a trade-off-driven perspective.

Therefore, representative studies were selected to cover different circuit architectures and practical hardware implementations, together with multi-objective optimization considerations. Instead of following a chronological review, the literature is evaluated in terms of five common design trade-offs: bandwidth versus group-delay flatness, NGD magnitude versus insertion loss, passive stability versus active tunability, compactness versus performance and high-frequency integration challenges.

The remainder of this paper is organized as follows. Section II reviews the mathematical foundations of NGD. Section III discusses the major design trade-offs identified in recent NGD research. Section IV presents a comparative synthesis of the surveyed literature, and Section V concludes the paper.

II. Theoretical Analysis of NGD Circuits

A strong mathematical foundation is necessary for the design of NGD circuits in order to guarantee stability and accomplish the required delay characteristics. Unlike conventional Non-Foster Circuits, which are prone to oscillation, modern analytical methodologies depend on robust filter-based topologies and resonance-based interference.

A. Maximally Flat NGD Response

In order to reduce group delay dispersion, a maximally flat NGD profile is realized using a transversal filter technique [4]. Figure 1 displays the design of the microwave transversal filter, with the coupling coefficients analytically determined such that the group delay stays constant throughout the operating band.

The network's transmission transfer function may be represented in terms of magnitude and phase as follows: $H(\omega) = |H(\omega)|e^{j\phi(\omega)}$, then the general definition of the group delay $\tau_g(\omega)$ is mathematically formulated as:

$$\tau_g(\omega) = -\frac{d\phi(\omega)}{d\omega} \tag{1}$$

A negative group delay response is strictly established within a specific localized frequency band when the condition $\tau_g(\omega) < 0$ is satisfied in (1).

To obtain a maximally flat profile at the specified center frequency (ω_0) and eliminate the dispersion-induced bulging, the higher-order derivatives of the group delay function with respect to the angular frequency should be zero, leading to the following synthesis criterion:

$$\left.\frac{d^n \tau_g(\omega)}{d\omega^n}\right|_{\omega=\omega_0} = 0 \tag{2}$$

for $n = 1, 2, \ldots, M$, where M represents the number of filter sections.

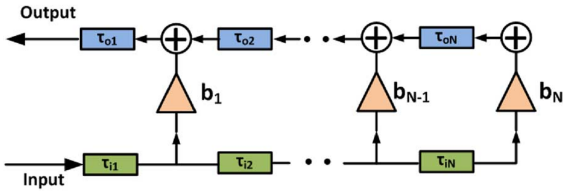


Fig. 1. General diagram of a microwave transversal filter for NGD [4].

The proposed method provides a uniform group delay flatness which is essential for maintaining the signal integrity in the current narrowband millimeter-wave applications. Although initially synthesized for lower microwave frequencies in [4], the mathematical formulation indicates that the exact internal coupling tap coefficients governed by (2) enforce a uniform transfer function response.

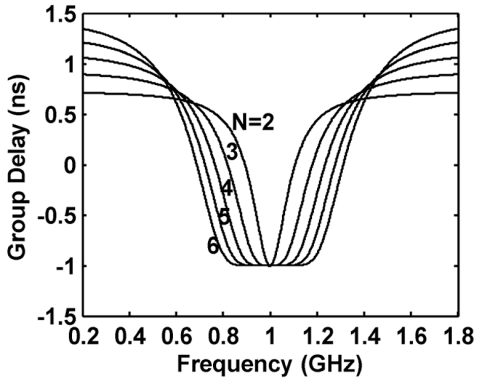


Fig. 2. Group delay profiles for various numbers of sections (M) in the transversal filter demonstrating maximally flat NGD responses [4].

The mathematical statement is validated by the analytical response profiles illustrated in Figure 2. The parallel horizontal responses show that increasing the operational parameter configuration of the filter results in a more uniform and flatter NGD band, ensuring that multi-frequency wave packets propagate without phase distortion across the target transmission spectrum.

B. Equal-Ripple NGD Response

For applications requiring wideband NGD performance, the equal-ripple method is more suitable [5]. By incorporating a coupling matrix with nonuniform Q-factors, a Chebyshev-like NGD response can be achieved. As illustrated in Figure 3, the use of nonuniform dissipation factors allows for a significant extension of the NGD bandwidth compared to the uniform case.

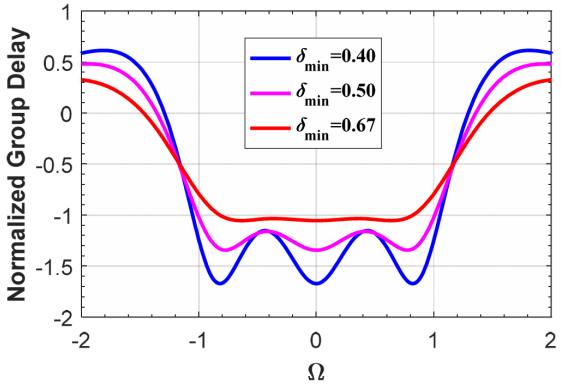


Fig. 3. Comparison of normalized group delays for filters with uniform and nonuniform Q-factors [5].

Mathematically, the modeled equal-ripple negative group delay function $\tau_g(\omega)$ can be modeled by scaling a baseline delay level τ_0 with an n-th-order Chebyshev polynomial T_n and a prescribed ripple factor ϵ:

$$\tau_g(\omega) = -\tau_0 \cdot \left[1 + \epsilon \cdot T_n\left(\frac{\omega}{\omega_c}\right)\right] \tag{3}$$

where ω_c denotes the operational corner frequency of the ripple-channel boundaries. While this approach governed by (3) introduces a controlled amount of ripple (ϵ) in the delay profile, it significantly extends the NGD bandwidth compared to maximally flat designs. The ripple level and the resulting bandwidth are primarily governed by the dissipation factors of the resonators.

III. State-of-the-Art Trends and Design Trade-offs in NGD Circuits

The design of NGD circuits has been constantly evolving to meet several different design objectives, rather than just a single performance metric. In passive, active, reconfigurable and high-frequency implementations, researchers have repeatedly met trade-offs between bandwidth, group-delay flatness, insertion loss, tunability, compactness, and operating frequency. Therefore,recent NGD studies have focused more on the balanced performance of these contradictory requirements. This section reviews these state-of-the-art developments systematically through the lens of these recurring design trade-offs to provide context for the current architectural boundaries.

A. Bandwidth versus Group Delay Flatness

Maintaining a flat group-delay response over a wide operational bandwidth remains one of the primary challenges in NGD circuit design. The authors in [4], proposed a maximally flat NGD circuit based on a microwave transversal filter. Whereas, The authors in [5] suggested an equal-ripple NGD filter to extend the usable bandwidth. Both approaches improve the delay characteristics, however at the expense of a fundamental trade-off between bandwidth and group-delay variations, larger group-delay variations usually occurring at higher bandwidths.

Recent studies have explored several design strategies to address this limitation. Reconfigurable flatness tuning using varactor-controlled elements was proposed in [10]. Broadband NGD performance was achieved using compact coupled-line structures in [3]. Bandwidth was further enhanced through impedance-engineered transmission-line structures in [9]. Group-delay fluctuations were reduced using NGD-assisted phase equalization in [15]. Overall, these studies indicate a shift from fixed flatness optimization toward adaptive and bandwidth-enhanced NGD designs.

B. NGD Magnitude versus Insertion Loss

One of the main challenges in NGD circuit design is to achieve large negative group delay responses with low insertion loss. Early passive implementations were able to achieve useful NGD characteristics, although this often came at the cost of significant signal attenuation. This established the well-known trade-off between NGD magnitude and insertion loss [7], [8].

More recently, researchers have explored active reconfigurable architectures and multi-band NGD designs to address this issue. Active compensation enables independent control of transmission gain and group delay. Multi-band design techniques improve NGD performance while maintaining relatively low insertion loss [2], [11], [12].These improvements suggest an increasing emphasis on reducing insertion loss with only a limited impact on NGD performance.

C. Passive Stability versus Active Tunability

The trade-off between passive stability and active tunability remains an important design consideration in NGD circuits design. Early passive implementations maintained simple and stable operation without the requirement of bias circuitry, however, the NGD characteristics remained fixed after fabrication, impacting their adaptability to different system requirements [3], [8].

Recent studies have tried to overcome this limitation by using active and reconfigurable architectures. Independent control of transmission gain and group delay, varactor-tunable, and reconfigurable non-Foster networks have enabled dynamic adjustment of the NGD characteristics for adaptive applications such as beamforming [2], [10], [13], despite the increased circuit complexity and biasing requirements. These advances suggest a shift from fixed passive implementations to more adaptive NGD systems.

D. Compactness versus Performance

As NGD circuits are increasingly employed in modern wireless systems, the reduction in circuit size while maintaining electrical performance has been a major challenge. The early distributed NGD structures occupied large circuit areas as the dimensions of their transmission lines were constrained by the operating wavelength. Attempts of minimizing the physical size often introduced impedance mismatch, parasitic coupling and bandwidth degradation, thus establishing a trade-off between compactness and performance [8].

Recent studies have explored several approaches to address this limitation, including compact transmission-line configurations, coupled-line structures, impedance-engineered designs, and NGD-assisted phase equalisation. These approaches reduce circuit size while preserving broadband operation or delay-equalization performance in high-frequency applications [1], [3], [9], [15]. Nevertheless, maintaining robust impedance matching and reducing fabrication sensitivity remain important challenges in highly compact NGD implementations.

E. High-Frequency NGD Implementations

Wireless networks are shifting to higher frequencies. Furthermore, beyond the conventional microwave working frequencies, NGD approaches are being investigated. However, implementing NGD circuits at high frequencies presents additional obstacles, including parasitic effects, greater losses, manufacturing tolerances, and implementation complexity [1], [13], [14].

Recent research have proved NGD approaches in high-frequency delay shifters, beamforming structures, and specialized millimeter-wave media, which support their expansion to higher-frequency operation [1], [13], [14]. These results point to a steady trend away from typical microwave frequency applications and toward more integrated high-frequency systems, yet fully optimised and scalable millimeter-wave implementations are still being researched.

F. Research Gaps and Future Directions

The comparative analysis conducted in this survey identifies several recurring research gaps in current NGD circuit designs. First, achieving a wide operational bandwidth while maintaining highly uniform group-delay flatness remains a major challenge, since most reported architectures improve one characteristic at the expense of the other [3]–[5], [9], [10], [15]. Second, although recent studies have reduced insertion loss using active and multi-band techniques, simultaneously achieving large NGD magnitude, low attenuation, and broad bandwidth remains difficult [2], [7], [8], [11], [12].

Active and reconfigurable architectures have also improved design flexibility. However, they generally require additional bias circuitry, higher power consumption, and greater implementation complexity [2], [10], [13]. Likewise, compact transmission-line and impedance-engineered techniques have significantly reduced circuit size, but maintaining electrical

performance under aggressive miniaturization and fabrication tolerances remains challenging [1], [3], [9], [15]. Although recent studies have extended NGD techniques toward high-frequency implementations, fully optimized and scalable millimeter-wave NGD systems are still relatively limited [1], [13], [14].

Future research is expected to focus on hybrid NGD architectures that combine passive phase equalization, active compensation, and reconfigurable tuning to simultaneously improve bandwidth, group-delay flatness, insertion loss, compactness, and high-frequency integration.

IV. Comparative Trade-Off Analysis

To summarize the evolution of NGD design methodologies, Table I compares representative studies that address each major design trade-off identified in the literature. Here, the earlier works refer to representative studies that first established a particular design trade-off, while the recent works represent subsequent efforts aimed at mitigating these limitations through improved design strategies including bandwidth enhancement, insertion-loss reduction, reconfigurable tuning, compact implementations, and high-frequency integration. The remaining challenges are derived from the recurring limitations found in the surveyed literature. They highlight the research directions that are not yet fully explored.

TABLE I
Vertical NGD Design Trade-Off Evolution

Design Trade-Off	**Earlier**	**Recent**	**Remaining Challenge**
Bandwidth vs. Flatness	[4], [5]	[3], [9], [10], [15]	Wideband flat NGD
Magnitude vs. Loss	[7], [8]	[2], [11], [12]	Large NGD, low loss
Stability vs. Tunability	[3], [8]	[2], [10], [13]	Tunability with stability
Compactness vs. Perf.	[8]	[1], [3], [9], [15]	Compact, high-perf.
High-Freq. Implement.	[1]	[13], [14]	Scalable mmWave NGD

V. Conclusion

This paper presented a trade-off-driven review of state-of-the-art NGD circuit design methodologies. Through a comparative analysis, the surveyed literature was examined using five recurring design trade-offs: bandwidth versus group-delay flatness, NGD magnitude versus insertion loss, passive stability versus active tunability, compactness versus performance, and high-frequency NGD implementations.

The analysis showed that significant progress has been achieved through active compensation, reconfigurable tuning, compact transmission-line structures, and high-frequency implementations. However, no single architecture simultaneously satisfies all performance requirements, indicating that NGD circuit design remains a multi-objective optimization problem.

Future research is expected to focus on hybrid NGD architectures that jointly improve bandwidth, group-delay flatness, insertion loss, tunability, compactness, and high-frequency integration for next-generation wireless communication systems.

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